University of Tobruk
Science Faculty
Physics Department

Second Year
Division of Medical Physics
Third Semester

Final Examination
Electricity and Magnetism I
Date:07 -02-2026

Q1: Define the following terms: (20 Marks)

a) **Radius of a charged particle moving in circular orbit in magnetic field**
b) **Total induced emf in a coil**
c) **Faraday's law of induction**
d) **Lenz's law**

Q2: Write down the four ways by which an emf can be induced in a circuit? (10 Marks)

Q3: Prove that magnetic field on the axis of a circular current loop is given by: (10 Marks)

$$B_x = \frac{\mu_0 \ I R^2}{2(X^2 + R^2)^{3/2}}$$

Q4: Long straight copper wire 0.127 cm in radians and magnetic field B at the surface of the wire 0.30T. Determine the current in the wire?

(10 Marks)

Q5: Two straight wire A and B carrying currents of 3 and 5 amper respectively in opposite direction. The wires are separated by a distance of 6 cm and experiences a force of 10 N. Determine the length of the wire?

(10 Marks)

Q6: A coil of 20 turns and radius 15 cm surrounds a long solenoid of radius 5cm with a turn $1x10^3$ turns/meter. The current in the solenoid varies with time as $I = (4\ A)\sin(150\ \pi t)$. Find the induced emf in the solenoid coil as a function of time.

(10 Marks)

Good Luck For All of You
Prepared by: Hanan Hussein

University of Tobruk
Science Faculty
Physics Department

Second Year | **Midterm Exam**

Division of Medical Physics | **Electricity and Magnetism I**

Fourth Semester | **Date: 02-12-2025**

Question 1: (15 Marks)

A: Define the following terms: **Electric field – Angular speed – Magnetic flux - The Solenoid**

B: Give an expression for the followings laws: **Gauss's law - Amper's law - Biot-Savart Law**

C: State the magnetic field lines properties?

Question 2: (15 Marks)

A: Proton moving at 4×10^6 m/s through a magnetic field of 1.70 T experiences a magnetic force of magnitude 8.20×10^{-13} N. What is the angle between the proton velocity and the field.

B: An electron travels with a speed of 5×10^6 m/s then enters a magnetic field of 0.3 T. What is

(a) The magnitude of the magnetic force on an electron and
(b) Its acceleration

C: A proton travels through perpendicular electric and magnetic fields. The electric field is 1.5×10^4 N/C and the magnetic field is B = 0.025 T. The proton passes through the fields undeflected. Determine the proton's velocity.

Good Luck For All of You

University of Tobruk
Faculty of Science
Department of Physics

Second Year (Medical Physics)
Fourth semester
Time: 2 hours

Final Examination
Electricity and Magnetism II
Date: 09 -06-2026

Question 1: Define the following terms: (18 Marks)

a) **Kirchhoff's Loop Rule for RL Circuits**
b) **The angular frequency of oscillation**
c) **Root Mean Square (rms)**
d) **Impedance of parallel RLC circuits**
e) **Resonant frequency**
f) **Average Power**

Question 2: Why does the current in a circuit not immediately jump to its maximum value ϵ/R when a switch is closed? (10 Marks)

Question 3: Describe oscillations in an LC Circuit? (10 Marks)

Question 4: An 5μF capacitor is connected to the terminals of a 50Hz AC source . Find the capacitive reactance. (10 Marks)

Question 5: A series RLC AC circuit has $R = 120\Omega, L = 1.2\ H, C = 3.5\mu F$, $\boldsymbol{\omega = 377\ Sec^{-1}}$ and $Vmax = 100V$.

a) Determine the inductive reactance, the capacitive reactance, and the impedance of the circuit.
b) Find the maximum current in the circuit.
c) Find the phase angle between the current and voltage. (12 Marks)

Question 6: Consider a series RLC circuit for which $\boldsymbol{\omega = 1000\ Sec^{-1}}$, $\boldsymbol{L = 20mH}$. Determine the value of the capacitance for which the current is a maximum. (10 Marks)

----------The End f the Exam----------

Good Luck For All of You

University of Tobruk
Faculty of Science
Department of Physics

Second Year (Medical Physics)
Fourth semester
Time: 2 hours

Midterm Exam
Subject :Electricity and Magnetism II
Date: 07-05-2026

Question 1: Fill in the gaps with suitable term:

a) A self-induced emf is always proportional ..... time rate .....
b) The inductance of the coil that depends on the ..... and other .....
c) Inductance = ...and its unit is ..... Henry .....
d) The time interval required for the current in the circuit to reach = ...to it final value ε\R.
e) Part of the energy supplied by the battery appears as ..... in the resistor, while the remaining energy is stored in the ..... of the inductor
f) We define the mutual inductance M_{12} of coil 2 with respect to coil 1 as .....
g) The total energy stored in the inductor U .....while the energy U in the circuit is stored in the electric field of capacitor is equal .....
h) Write an expression showing how the inductor affects the current .....

Question 2: Calculate the inductance of an air-core solenoid containing **200 turns**, given that the length of the solenoid is **20 cm** and its cross-sectional area is **4 cm²**.

Question 3: Calculate the resistance (R) of the circuit if the time constant (τ) is 0.005 seconds and the inductance (L) is 30 mH.

R, L

Good Luck For All of You